

## Task 2: Conceptual Design of End-Effector for Leafy Green Harvesting

Vertical farming systems present unique challenges for robotic harvesting. Crops are grown in stacked trays with minimal spacing, high humidity, and sensitive leaves that are easily bruised or torn. Traditional rigid grippers or pinch-based mechanisms often fail in such environments due to excessive contact force, lack of adaptability to varying leaf sizes, and poor performance under wet conditions.

The goal of this design is to create an end-effector that can gently harvest leafy greens while minimizing plant disturbance, maintaining hygiene, and operating reliably in tight vertical farm layouts.

### 2. Proposed Concept Overview

The proposed solution is a Vacuum-Assisted Soft Sleeve End-Effector with an Integrated Micro-Cutting Mechanism. This design combines soft robotics principles with controlled suction and a precise cutting action.

The end-effector consists of a flexible, food-grade silicone sleeve that envelopes the target leaf or stem. A low-pressure vacuum gently pulls the leaf into the sleeve, stabilizing it without squeezing. Once positioned, a small internal micro-cutter performs a clean cut near the stem base, allowing the leaf to be harvested safely.

### 3. Key Design Components

- **Soft Silicone Sleeve:** Provides compliance and adapts to varying leaf sizes and shapes, reducing damage risk.
- **Vacuum Inlet:** Generates gentle suction to guide and stabilize the leaf without applying crushing forces.
- **Micro-Cutter:** A small electrically actuated blade positioned inside the sleeve for clean stem separation.
- **Force and Pressure Feedback:** Sensors monitor suction force to prevent over-pulling or tearing.
- **Hygienic Materials:** All crop-contact surfaces are food-safe, waterproof, and easy to clean.

### 4. Why This Design Works in Vertical Farming

This design is particularly suited for vertical farming due to its minimal footprint and soft interaction strategy. The sleeve-based approach allows the end-effector to operate in confined spaces without requiring wide finger opening. Vacuum stabilization avoids lateral forces that could disturb neighboring plants.

The cut-based harvesting method ensures repeatable detachment regardless of leaf thickness or moisture, which is a common issue in humid indoor farms. Additionally, the

modular nature of the design allows easy integration with robotic arms such as UR5 or Panda using standard tool flanges.

## **5. Why Not Traditional Alternatives**

**Rigid Parallel Grippers:** These apply concentrated forces at contact points, often bruising or tearing leaves.

**Pure Vacuum Suction Cups:** While gentle, they struggle with wet or textured leaves and may fail to detach the crop cleanly.

**Scissor or Blade-Only Tools:** These require precise alignment and offer no stabilization, leading to unreliable cuts.

In contrast, the proposed hybrid approach combines stabilization and cutting, addressing the shortcomings of each individual method.

## **6. Feasibility and Commercial Readiness**

The proposed end-effector is feasible with existing technologies. Pneumatic vacuum systems, soft silicone molding, and compact cutting actuators are already commercially available. The design is cost-effective, scalable, and suitable for mass production.

From a maintenance perspective, the sleeve and blade modules can be easily replaced or sterilized, making the system practical for continuous agricultural operation.

## **7. Key Innovation**

The primary innovation lies in the integration of a soft vacuum-guided sleeve with an internal cutting mechanism. Rather than relying on grip force or suction alone, the system intelligently combines gentle stabilization with precise cutting, achieving high reliability and minimal crop damage. This integrated approach is not commonly seen in existing agricultural harvesting tools.

Vacuum-Assisted Soft Sleeve Gripper with Internal Micro-Cutter

